

Experiment 5.2

Step Response of the System

Program:

```
% Step Response of the System

clc;

clear;

close all;

% Sample index

n = 0:10;

% Unit step input

x = ones(1,length(n));

% System:  $y(n)=x(n)+0.5x(n-1)$  b = [1 0.5];

a = [1];

% Step response

y = filter(b,a,x);

% Display result

disp('Step Response y(n):'); disp(y);

% Plot

figure;

subplot(2,1,1);

stem(n,x,'filled');

grid on;

title('Unit Step Input u(n)'); xlabel('n');

ylabel('Amplitude');

subplot(2,1,2);

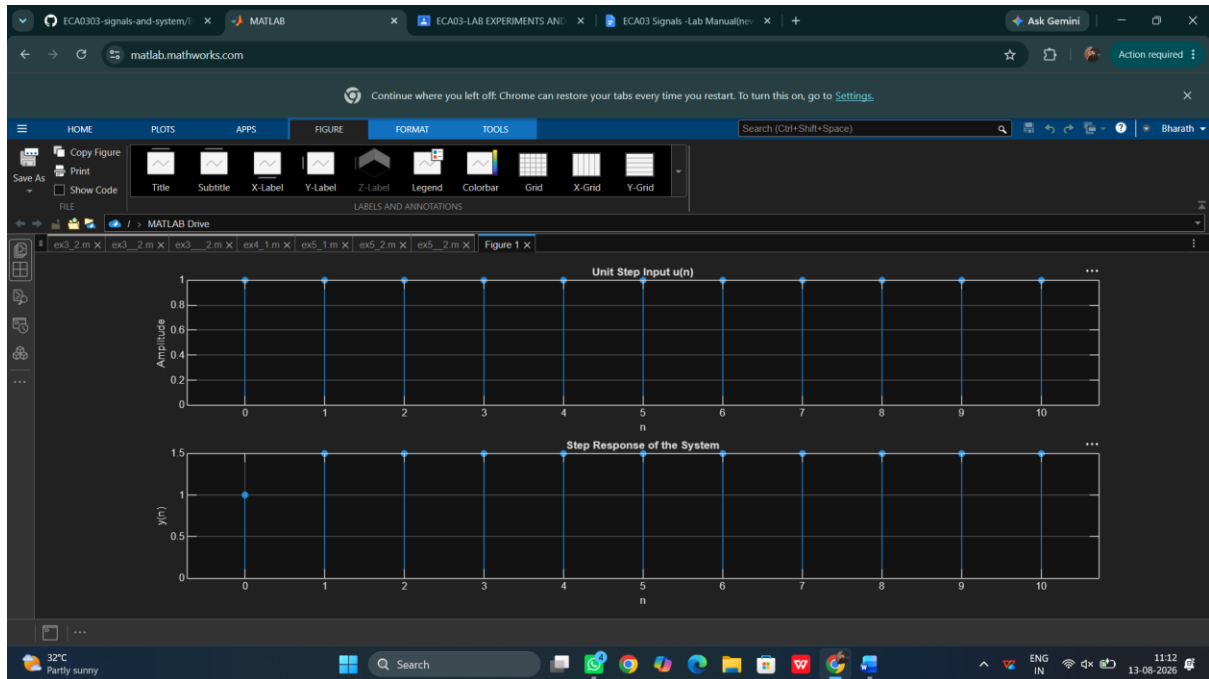
stem(n,y,'filled');
```

grid on;

```
title('Step Response of the System'); xlabel('n');
```

```
ylabel('y(n)');
```

Output Waveform:



Result:

The step response of the discrete-time system $y(n)=x(n)+0.5x(n-1)$ was obtained by applying a unit step input. The output reached a steady-state value of **1.5**, demonstrating the response of the system to a constant input signal.